

When Formalisms Stop Recovering: Minimal Structural Sufficiency and the Measurement Boundary in Quantum Mechanics

Quantum mechanics is one of the most successful predictive frameworks ever constructed. Its formal apparatus—Hilbert spaces, operators, unitary evolution, and probabilistic measurement rules—has been validated to extraordinary precision across countless experiments.

And yet, at its conceptual core, quantum theory exhibits a persistent and widely acknowledged instability:

measurement cannot be derived smoothly from unitary dynamics.

This is not a fringe claim.

It is a structural fact recognized across interpretations.

The mathematical formalism functions perfectly—until it suddenly does not.

The Empirical Pattern

As long as quantum evolution remains isolated, closed, and unitary, the theory behaves coherently. Superposition, interference, and entanglement evolve predictably under Schrödinger dynamics.

However, when a measurement boundary is introduced—when an interaction becomes irreversible, macroscopic, or informationally constrained—the formalism changes its character.

The system no longer evolves.

It *selects*.

This selection is not gradual.

It is not derived.

And it is not recoverable.

Once a measurement outcome is registered, the prior superposition cannot be reconstructed by reversing the formal evolution. The collapse is abrupt, asymmetric, and irreversible.

This transition is empirical.
It does not depend on interpretation.
And it remains structurally unexplained.

The Explanatory Gap

Standard accounts attempt to explain measurement collapse through:

- decoherence
- environmental entanglement
- epistemic updating
- many-worlds branching
- hidden variables

Each approach successfully describes *aspects* of the phenomenon. None explains why the formalism itself must change regime at the measurement boundary.

In particular, decoherence explains why interference terms become unobservable—but not why a single outcome is selected rather than all remaining equally real or recoverable.

What is missing is not a mechanism inside the formalism.

What is missing is an account of **what must remain invariant for quantum coherence to persist**, and **what is irreversibly lost at measurement**.

A Boundary Phenomenon

From a structural perspective, quantum measurement behaves like a phase transition rather than a continuous dynamical process.

Above a certain boundary, quantum systems exhibit:

- reversible evolution
- linear superposition
- conserved norm
- recoverable phase information

Below that boundary, the system exhibits:

- outcome selection

- irreversibility
- loss of interference
- non-unitary update rules

The transition is not governed by time, energy, or coupling strength alone. It is governed by a *structural boundary condition* imposed by the measurement context.

This suggests that quantum theory is not globally uniform.

It operates in **regimes**.

Minimal Structural Sufficiency

We formalize this boundary in terms of **minimal structural sufficiency**, denoted $F_{\min}$.

In the quantum domain, $F_{\min}$ corresponds to the minimal structural conditions required for coherent unitary evolution to remain physically meaningful.

These conditions include (but are not reducible to):

- reversibility
- isolation
- phase coherence
- informational non-leakage

As long as these conditions are preserved, quantum states evolve coherently. Once they are violated, the formalism itself changes character.

Key properties of $F_{\min}$:

- **Non-distributable**: coherence cannot be partially preserved; once broken, it is broken.
- **Non-recoverable**: post-measurement states cannot be unitarily reversed to restore prior superpositions.
- **Boundary-induced**: violation occurs at interfaces (measurement, environment, macroscopic registration).
- **Formalism-relative**: collapse emerges not from dynamics, but from structural conditions of description.

Measurement collapse occurs not because “too much interaction happened,” but because $F_{\min}$ was violated.

Collapse as Throughput Failure

Operationally, the violation of $F_{\min}$ manifests as a collapse in informational throughput, denoted Ψ .

Above $F_{\min}$:

- quantum information is preserved
- amplitudes interfere
- histories remain coherent
- predictions are unitary

Below $F_{\min}$:

- information becomes irreversibly classical
- amplitudes no longer interfere
- histories decohere into outcomes
- predictions require probabilistic update rules

This collapse is abrupt and asymmetric. No continuous interpolation exists between unitary evolution and outcome selection.

This explains a long-standing fact in quantum foundations:

*Measurement is not an extension of dynamics.
It is a boundary where dynamics cease to apply.*

Why Formalisms Stop Recovering

Attempts to “repair” measurement within the formalism—by extending Hilbert spaces, adding environments, or multiplying worlds—fail to eliminate the boundary.

They merely relocate it.

The reason is structural: once $F_{\min}$ is violated, no internal operation of the formalism can restore coherence, because the conditions that made coherence meaningful are no longer present.

This is why collapse cannot be derived from unitary evolution alone. It is not missing mathematics; it is missing a boundary condition.

Relation to Measurement Formalisms

From this perspective, the various quantum interpretations can be reclassified:

- Decoherence explains *how* coherence is lost, not *why* loss is irreversible.
- Many-worlds preserves unitary evolution by abandoning outcome uniqueness.
- Collapse models introduce explicit boundary rules.
- Realist interpretations treat the boundary as ontologically significant.

What unifies them is the recognition—often implicit—that **unitary evolution is structurally insufficient at the measurement boundary**.

Why This Matters

Quantum theory does not fail because it is incorrect.
It fails because it is *locally incomplete* at its boundaries.

Recognizing minimal structural sufficiency reframes the measurement problem from a metaphysical puzzle into a structural one:

what conditions must hold for quantum coherence to remain valid, and what happens when they no longer do?

This perspective does not require new dynamics, new particles, or new interpretations.

It requires recognizing that **formal validity has limits**.

Conclusion

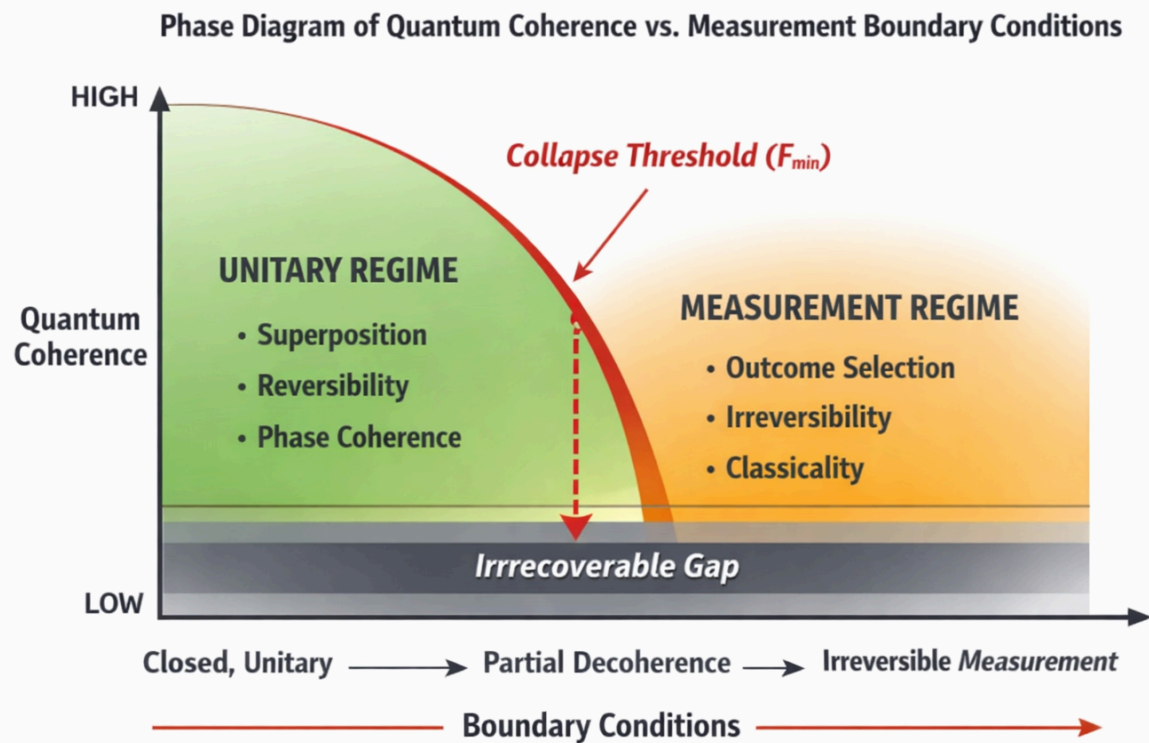
Quantum systems do not collapse because something mysterious happens in time.
They collapse because the structural conditions required for unitary coherence no longer exist.

Measurement is not a dynamical process inside quantum mechanics.
It is the boundary where quantum mechanics stops applying.

You do not need to believe this interpretation.

You only need to notice when the formalism stops recovering.

Quantum coherence vs Measurement Diagram



Appendix A

Formal Remarks on Irreversibility and Non-Distributivity

A.1 Irreversibility as a Structural Property

Irreversibility, as used in this work, does not refer to thermodynamic entropy production per se, nor to practical difficulty of reversal. It denotes a **structural asymmetry**: once the system crosses a minimal sufficiency boundary $F_{\min}$, its original operational regime cannot be restored by any admissible internal transformation.

Formally, let $\mathcal{S}$ be a system with state space Ω and operational throughput Ψ . Let $\Omega_{\text{stable}} \subset \Omega$ denote the region where coherent operation is sustained.

There exists a boundary $F_{\min}$ such that:

$$\Psi(\omega) \geq \Psi_{\min} \quad \forall \omega \in \Omega_{\text{stable}}$$

Crossing below this boundary induces a transition:

$$\omega \in \Omega_{\text{stable}} \longrightarrow \omega' \in \Omega_{\text{collapsed}}$$

with the defining property:

$$\nexists T_{\text{internal}} : \Omega_{\text{collapsed}} \rightarrow \Omega_{\text{stable}}$$

where T_{internal} is any transformation available to the system without external reconstitution.

This irreversibility is therefore **topological**, not energetic: the phase space itself is restructured. In quantum measurement, this corresponds to the loss of coherent superposition. In pruned neural networks, it corresponds to loss of global representational modes. In both cases, retraining or unitary evolution alone cannot restore the original regime once $F_{\min}$ is violated.

A.2 Non-Distributivity of Minimal Sufficiency

A central result of the framework is that $F_{\min}$ is **non-distributive**.

By non-distributive we mean that minimal sufficiency cannot be expressed as the sum, average, or local aggregation of sub-components. That is:

$$F_{\min} \neq \sum_i f_i \quad \text{and} \quad F_{\min} \neq \bigcup_i f_i$$

where f_i are locally sufficient substructures.

Minimal sufficiency is instead a **global relational property**: it depends on the preservation of specific structural invariants (connectivity, spectral modes, boundary-induced constraints) that are destroyed when distributed pruning exceeds a critical configuration.

This explains why:

- Two systems with identical sparsity ratios can exhibit radically different performance.
- Local preservation of high-magnitude components does not guarantee global functionality.
- Post-hoc redistribution cannot reconstruct lost coherence.

Once a non-distributive structure is breached, no amount of local reinforcement restores it, because the invariant itself no longer exists in the system's configuration space.

A.3 Relation to ISSP and Spectral Collapse

Within the ISSP (Induced Structural Spectral Projection) framework, the transition at $F_{\min}$ corresponds to the loss of dominant eigenmodes of the system's effective operator.

Let $\mathcal{L}$ be the induced connectivity or interaction operator. Stable operation requires the persistence of a low-dimensional dominant subspace:

$$\lambda_1, \lambda_2, \dots, \lambda_k \gg \lambda_{k+1}$$

Pruning or boundary enforcement that preserves this spectral core maintains Ψ above threshold. Violation of $F_{\min}$ corresponds to **spectral rank collapse**, where dominant modes are eliminated rather than merely attenuated.

This collapse is irreversible because the operator's spectrum itself has changed — not merely its coefficients.

A.4 Consequence

Irreversibility and non-distributivity jointly imply that system survival is not a matter of gradual degradation, but of **structural phase transition**.

Systems do not fail because they lose “too much”.

They fail because they lose **the wrong thing**.

Once the minimal sufficient structure is breached, collapse is not a bug, nor a limitation of optimization — it is a logical consequence of the system’s architecture.